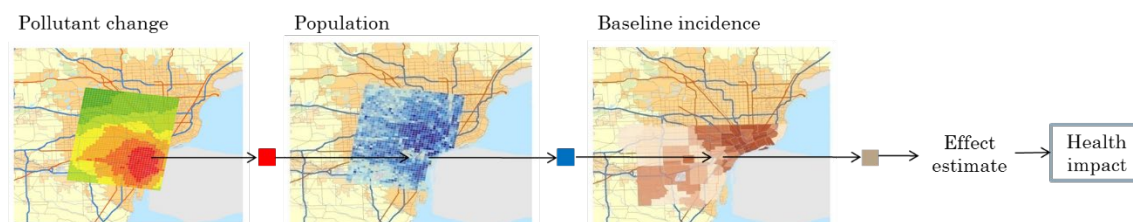


increment between the baseline scenario and the control scenario. This increment and a gridded population dataset are then used in health impact functions to calculate the change in health incidence that would result from this change in pollution. These functions are based on epidemiological studies and can be selected by the user. Typically, these health incidence results show the decrease in health incidence (e.g., the decrease in asthma, bronchitis, mortality) due to a decrease in pollution. In BenMAP-CE, the selected health impact functions are stored in configurations, which can be re-used over and over again.

Steps to Calculating Health Impacts

$$\Delta Y = Y_o (1 - e^{-\beta \Delta PM}) * Pop$$



6.2 Create a Health Impact Configuration

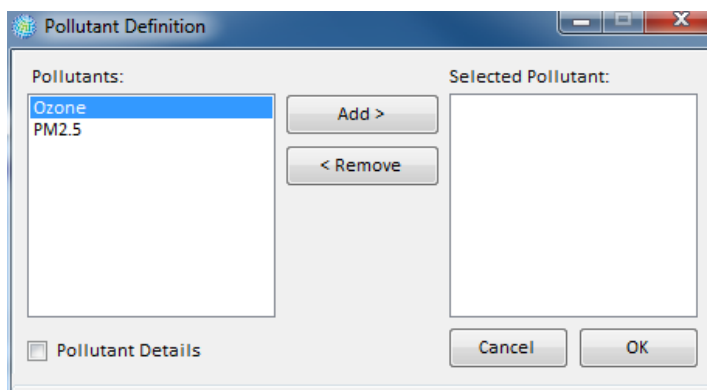
There are three major steps to creating a new configuration. First, select a **Pollutant** and specify the **Source of Air Quality Data** for the **Baseline** and the **Control** layers using the BenMAP-CE tree menu on the left side of the main window. Next, choose the **Population Dataset** and **Health Impact Functions** that you want to use in your analysis. Finally, using the **Advanced** button on the **Health Impact Functions** window, specify whether BenMAP-CE will run in **Point Mode** or perform a Monte-Carlo analysis (default setting). (See **Section 6.2.3 Advanced Configuration Settings** for additional detail.)

Note: As you move through the analysis steps, the BenMAP-CE tree menu will update its “**stoplight**” colors to reflect progress. **Yellow** indicates an operation has not yet started. **Green** indicates that an operation has been successfully completed. **Red** indicates that an operation completed, but you may need to re-run this step.

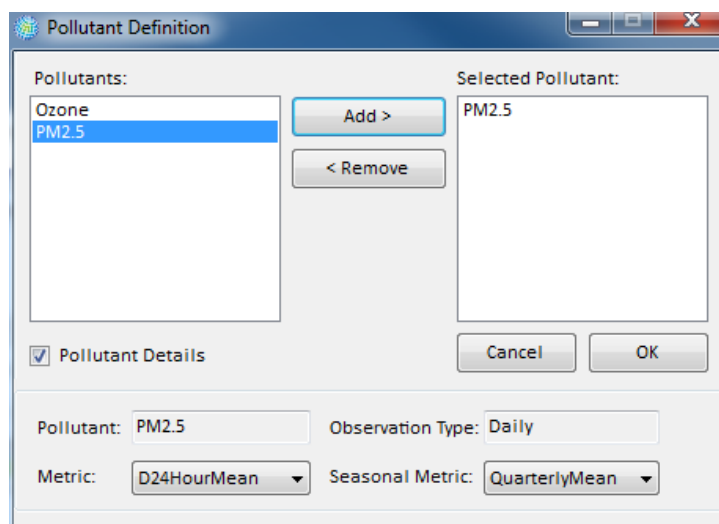
6.2.1 Air Quality Surfaces – Pollutants and Air Quality Grids

Using the BenMAP-CE tree menu, specify the pollutant(s) of interest. Click the **Pollutant** tree menu item to open the **Pollutant Definition** window. Select the pollutant(s) of interest by clicking on an item in the **Pollutants** box (left side) and then clicking the **Add** button to move the pollutant(s) to the **Selected Pollutant** box. To deselect a pollutant and remove it from the **Selected Pollutant** box, click on the item in

the **Selected Pollutant** box and click the **Remove** button. You can also add a pollutant by dragging it from the left-hand box to the right-hand box.

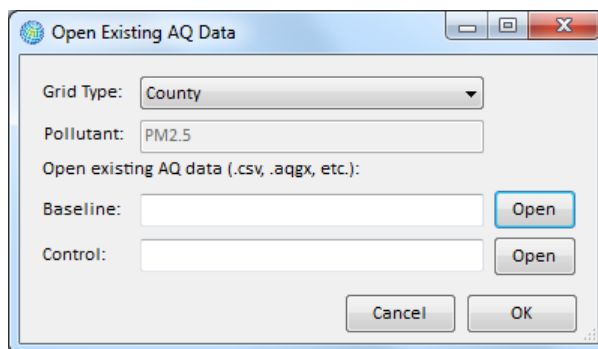


If you want to view or modify any pollutant details, click the box next to *Pollutant Details* and the window will expand to display details and options for the highlighted **Pollutant** (left box). Note that the details are based on the defined **Pollutant Definitions** (see Chapter 4: Loading Data).



Once your pollutant is selected, click **OK** to close the form. The stoplight for **Pollutant** will change from yellow to green when the operation is successfully completed.

Next, select air quality grids for the **Baseline** file and **Control** file. You may choose existing air quality grids by double-clicking the **Source of Air Quality Data** from the tree menu. The **Open Existing AQ Data** window will be displayed. Select air quality grids, designated with an '.aqgx' extension, for **Baseline** and **Control**. To create a new air quality grid, follow the steps outlined in Chapter 5: Creating Air Quality Grids.



The **Baseline** file contains the air quality metrics for the scenario assumed to occur without any change in policy. The **Control** file specifies the air quality metrics assuming that some type of policy or change has been implemented. The air quality grids should be for the same pollutant, and should also be based on the same **Grid Type**.

If you choose a particular **Grid Type** (e.g., *County*) for the **Baseline** file, then the same grid type must be used in the **Control** file. Conversely, it would not be possible to use *County* grid-type in the **Baseline** and a *Tract* grid-type in the **Control** file.

Fundamental Concept – Baseline and Control Scenarios

BenMAP-CE requires both a **Baseline Scenario** air quality surface and **Control Scenario** air quality surface to estimate the effects of a change in air quality (**Delta**). Both baseline and scenario air quality surfaces are required regardless of whether you use modeling data or monitoring data.

- The **Baseline Scenario** characterizes the air quality levels observed or expected in the absence of the policy change you are evaluating. The baseline is sometimes referred to as “Business as Usual.” The Baseline scenario is usually considered to be the reference scenario against which to compare a potential scenario characterized by the implementation of regulations.
- The **Control Scenario** in BenMAP is the scenario in which emissions from one or more source sectors are changed (increased or decreased) from the Baseline scenario. The Control scenario usually represents expected air quality levels after a new regulation or set of regulations has been implemented.

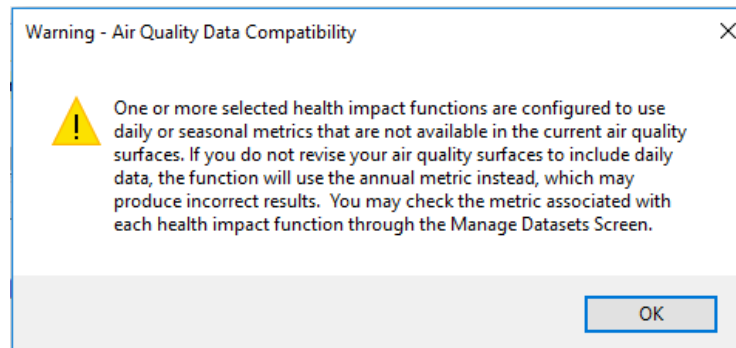
The air quality **Delta** is the change in air pollution between the Baseline air quality grid and the Control air quality grid (Baseline minus Control). BenMAP-CE uses the air quality **Delta** as the input to the health impact function.

The **Pollutant** specified in the air quality grids determines the suite of **Health Impact Functions** available for the configuration. Only functions associated with the specified **Pollutant** will be available for the configuration. Furthermore, if only certain **Metrics** associated with the pollutant are present in one (or both) of the air quality grids (see

the information in Chapter 3 on monitor and model data formats for more information on how this can occur) **Health Impact Functions** associated with those **Metrics** will show a notification that the air quality surface does not provide the metric specified in the health impact function and the annual metric will be used instead.

Warning

BenMAP-CE will evaluate the compatibility between provided air quality metrics and health impact functions. For example, if the air quality surface provides an annual pollutant metric for a health impact function that uses a daily pollutant metric, BenMAP-CE will provide a warning message (shown below).



Once the **Baseline** and **Control** files are selected, click **OK** to close the form. The stoplights for each of these under **Source of Air Quality Data** will change from yellow to green when the operation is successfully completed.

6.2.2 Estimate Health Impacts – Population and Health Impact Functions

The second step in creating a health impact configuration is to select the **Population Dataset**, **Population Year** and **Health Impact Functions**. If you want to open an existing configuration, double-click the **Estimate Health Impacts** tree menu item and select the file to open. Follow instructions from **Section 3.1.2 Estimate Health Impacts**.

To continue creating a new configuration, double-click **Population Dataset** in the tree menu. Here, you can choose the **Population Dataset** and **Population Year** that will be used in the analysis. The values in the drop-down list for **Population Year** depend on the range of values in your **Population Dataset**. (See Chapter 4: Loading Data.)

